

ROOTMESH · DESIGN DECISIONS REGISTER

RootMesh design decisions register

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	RMS-DEC-001	Version	0.1
Status	Draft	Date	2026-10-01
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/rootmesh		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the review note and decision records, the design for construction (RMS-DDR-003), items to confirm when parts are bought, and the value-engineering summary

RootMesh design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in the review note (`docs/REVIEW.md`); this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design for construction: two-part head with O-ring and screws, board guides, fin in two glued halves with collar and spigot, flush temperature probe, wedge tip and one-piece electrodes, cell holder, gland and vent moved, antenna clips, slot tool handle and stop fixings	Accept as recorded; or change any item	Accept: every change keeps the sensing depths, electrode geometry, cell depth, panel and antenna height	Every component (build plan sections 2 to 4)	RMS-DDR-003, P1 to P11
2	Fin print size: each half is 355 mm long	(a) print on a 300 mm class printer or through a print service; (b) split each half into two lengths with a glued lap joint	(a)	Fin halves (build plan section 3.4)	RMS-DDR-003, A1
3	Head bonded to the tube, so the stake is one piece below the cap	(a) bonded, as modelled; (b) a second O-ring and screws so the head comes off the tube	(a) for the prototype; revisit after a soak test at TRL 4	Steps 4 and 6	RMS-DDR-003, A2
4	Response to the cell temperature in bare, hot soil (R9, cell up to about 47 °C)	(A) a printed shade skirt round the head; (B) restate the R9 cell limit to the chosen cell's rated discharge and storage range, with charging still blocked above 45 °C	(B), after confirming the cell's rated range from its datasheet	None in the TRL 3 build; (A) would add a part to the head	RMS-DDR-002, P1
5	First users and region for co-design, which sets EU868 or US915	Market garden, orchard, university research farm or extension program	None yet (co-design partners to be picked per area later)	Antenna and radio band (build plan section 3.12, safety stop S7)	RMS-DDR-001, O1
6	Status light on the head	Keep a green LED, blinking briefly after each uplink and on a magnet or button wake only, with its charge added to the energy budget; or leave it out	Keep it, once its charge is added to RMS-CAL-001	Not in the TRL 3 build; would add a lens hole in the body	REVIEW.md, 2026-09-26, item 2
7	Gateway on a post in the hero render	Keep the pilot gateway indoors as decided and treat the post as a render	Keep it indoors; the post is a render layout	None	REVIEW.md, 2026-09-26, item 3

#	Decision needed	Options	Recommendation	Affects in the build	Source
		layout only; or show it indoors			

The separate solar cap proposed in the 2026-09-26 review (item 1 there) is now part of decision 1: RMS-DDR-003 adopts a separate cap, sealed by a face O-ring and two screws.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The moisture probe boards are 23 mm wide and 98 to 100 mm long, with their parts on one face within 16 x 3 mm at the top end	The fin's slots and electronics pockets are sized for them	RMS-DDR-003, P4
2	The temperature probe's sheath is 6 mm across and about 50 mm long, and its inner wires fit a 3.5 x 5 mm channel with the jacket stripped	The fin's bore and channel are sized for it	RMS-DDR-003, P5
3	The 14500 cell with its fuse and leads fits a 15 mm bore, 51 mm deep	The cell holder is sized for it	RMS-DDR-003, P7
4	The PVC pipe is 42 mm outside and 36 mm inside diameter	The head socket (43 mm) and fin spigot (35.4 mm) leave glue gaps of about 0.5 and 0.3 mm	RMS-DDR-003, P3
5	A 64 x 2 mm O-ring, an M8 gland for 2 to 4 mm cable with its locknut, and short M3 heat-set inserts for a 4.6 mm hole are available	The cap groove, gland boss and insert bosses are sized for them	RMS-DDR-003, P1 and P8
6	The solar panel is 70 x 50 mm or smaller, with its lead near the centre of its back	The cap recess is 71 x 51 mm with a lead hole at its centre	RMS-DDR-003, P1
7	The charger's temperature window is 0 to 45 °C	R7 and safety stops S3 to S5 depend on it	RMS-CAL-001, B6
8	The shaft collars fit 20 mm bar	The slot tool handle depends on them	RMS-DDR-003, P11

■ Value engineering

Value-engineering target: USD 300 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 281 for the pilot set of three stakes, one gateway and one slot tool (USD 19 under the target); USD 59 per stake against R12's USD 60. Main cost drivers and savings worth trying:

- The largest lines are the gateway (USD 90, about a third of the set), the controller board (USD 15 a stake, USD 45 for three), the antenna and feed (USD 7 a stake) and the seals (USD 5 a stake).
- Making the design constructable added USD 3.50 a stake (the two-part head, the fin halves, and the inserts, screws, epoxy and prototype board of line 16) and USD 4 on the slot tool (shaft collars and thumb screw); the set rose from USD 266.50 to USD 281.

- Savings worth trying: one gateway serves a whole farm, so the per-stake cost falls quickly in larger sets; the controller board's price should fall once the carrier board is designed at TRL 4; one slot tool serves every stake on a farm; O-rings, glands and inserts are much cheaper bought by the hundred.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D9: solar with a LiFePO4 cell, antenna at 1 m on the marker rod, The Things Network with an indoor gateway, the budget read as a pilot set, LiFePO4 chemistry, two moisture depths, modified hobby probes, Wio-E5 controller, 20 min interval	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	RMS-DDR-001
2026-09-25	Slot tool for firm soil; deeper cell checked and kept at 62 mm; pilot set includes the slot tool; field calibration note decided but on hold	Amish: "i accept all your recommendations, go with them across all repos."	RMS-DDR-002, A1 to A4
2026-09-30	Build plan format approved for all repos; open decisions go in this register, not in the build plan; the design is made physically buildable as the pictures are drawn	Amish: "this is the correct build plan ... this is a good quality document format. Extend this across all the other repos"; "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	RMS-BLD-001, RMS-DDR-003
2026-10-01	The budget is a value-engineering target, not a limit; budget_usd stays USD 300	Amish: "the budgets are a hypotheticalal control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	This register, Value engineering